

COMPARATIVE ANALYSIS

Q1. Object Detection Performance Comparison

Two object detection models are given:

- **Model A – Viola-Jones**
- **Model B – YOLO**

The results are:

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
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Model A	80	20	30
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Model B	90	15	20
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We need to calculate Precision and Recall and then compare both models.

(a) Calculate Precision and Recall

Model A – Viola-Jones

Given:

Precision

Recall

Therefore, Model A has:

- Precision = 80%
- Recall = 72.73%

Model B – YOLO

Given:

Precision

Recall

Therefore, Model B has:

- Precision = 85.71%
- Recall = 81.82%

Comparison

Model	Precision	Recall
Model A – Viola-Jones	80%	72.73%
Model B – YOLO	85.71%	81.82%

This means YOLO produces fewer false detections and also misses fewer actual objects.

Overall Performance

To further compare the models, we can calculate the F1-score, which combines precision and recall.

Final Conclusion

Model B (YOLO) provides better overall object detection performance.

It has:

- Higher precision: 85.71% vs 80%
- Higher recall: 81.82% vs 72.73%
- Higher F1-score: 83.72% vs 76.19%

Therefore, YOLO is better because it produces more correct detections and misses fewer objects compared with Viola-Jones.

Q2. Motion Estimation Error Analysis

The displacement errors for three consecutive frames are:

- Method A: 2, 3, 2 pixels
- Method B: 3, 5, 4 pixels

A smaller displacement error indicates more accurate motion estimation.

(a) Calculate the Average Displacement Error

Method A

Errors are:

The average error is:

Method B

Errors are:

The average error is:

(b) Compare the Two Methods

Method	Frame 1	Frame 2	Frame 3	Average Error
Method A	2 px	3 px	2 px	2.33 px
Method B	3 px	5 px	4 px	4 px

Method A has an average displacement error of only 2.33 pixels, while Method B has an average error of 4 pixels.

Since lower error means better accuracy:

Therefore, Method A is more accurate.

(c) Justification

Method A provides lower errors in all three frames:

- Frame 1: 2 px vs 3 px
- Frame 2: 3 px vs 5 px
- Frame 3: 2 px vs 4 px

The average error of Method A is also significantly lower.

Method B produces larger displacement errors, especially in the second frame where the error reaches 5 pixels.

Therefore, Method A provides more reliable and accurate motion estimation.

Final Conclusion

Method A should be selected because it has the lowest average displacement error of 2.33 pixels, compared with 4 pixels for Method B. A lower error indicates that the estimated motion is closer to the actual motion.

Q3. Depth Estimation Comparison

A stereo vision system and an SfM system estimate the depth of three objects.

Given Data

Object	Actual Depth	Stereo Vision	SfM
Object 1	2.0 m	2.0 m	2.5 m
Object 2	3.0 m	3.0 m	3.8 m
Object 3	4.0 m	4.0 m	5.0 m

We need to calculate the absolute error for each object and then calculate the average absolute error.

Formula

The absolute value is used because we are interested in the magnitude of the error, not whether the estimation is higher or lower than the actual value.

(a) Calculate Absolute Error

Object 1

Actual depth:

Stereo Vision

SfM

Object 2

Actual depth:

Stereo Vision

SfM

Object 3

Actual depth:

Stereo Vision

SfM

Absolute Error Table

Object	Actual Depth	Stereo Estimated	Stereo Error	SfM Estimated	SfM Error
Object 1	2.0 m	2.0 m	0 m	2.5 m	0.5 m
Object 2	3.0 m	3.0 m	0 m	3.8 m	0.8 m
Object 3	4.0 m	4.0 m	0 m	5.0 m	1.0 m

(b) Calculate Average Absolute Error

Stereo Vision

The errors are:

Therefore:

SfM

The errors are:

Therefore:

(c) Compare the Methods

Method	Average Absolute Error
Stereo Vision	0 m
SfM	0.77 m

The average error of Stereo Vision is 0 m, which means its estimated depths exactly match the given actual depths for all three objects.

SfM has an average absolute error of approximately 0.77 m. Its estimated depth is consistently higher than the actual depth.

For example:

- Object 1 is overestimated by 0.5 m
- Object 2 is overestimated by 0.8 m
- Object 3 is overestimated by 1.0 m

Since lower error indicates better estimation accuracy:

Therefore, Stereo Vision provides more accurate depth estimation for the given data.

Final Conclusion

Stereo Vision has an average absolute error of 0 m, while SfM has an average absolute error of 0.77 m. Therefore, Stereo Vision provides the most accurate depth estimation for these three objects.